

Bootstrap and Cross-Validation for Model Uncertainty in Climate Prediction

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Abstract

We evaluate the predictive performance and uncertainty of classical and modern machine learning models for global temperature anomaly forecasting using NASA's GISTEMP v4 dataset (1880–2025). Particular attention is given to the challenges of temporal dependence and data leakage in time series modeling, as well as the differences between model prediction and interpretability. To address these, we implement a forward-chaining cross-validation framework and a 10-year holdout evaluation, ensuring all models are assessed on identical chronological partitions. Five modeling approaches are compared: SARIMA as a classical baseline, Elastic Net regression, Random Forest, feed-forward neural networks using a sliding window architecture, and Bayesian Structural Time Series (BSTS). Model performance is evaluated using cross-validated mean squared error and holdout forecast accuracy, while bootstrap resampling is used to construct confidence intervals for pairwise performance differences. Results indicate that SARIMA provides a strong and stable baseline, particularly for short-term forecasting. BSTS achieves the best overall long-term performance, effectively capturing the upward trend in temperature anomalies while maintaining interpretability through latent component decomposition. Random Forest demonstrates competitive performance but fails to extrapolate beyond the training distribution. Neural networks exhibit high variance and instability in long-horizon forecasts, while Elastic Net underperforms due to its limited ability to capture nonlinear dynamics and long-term trends. Model performance was also evaluated separately for the Northern and Southern Hemispheres within the context of BSTS, revealing differences consistent with data density and thermal dynamics. These findings highlight the importance of model structure and evaluation design in climate time series forecasting, and suggest that while machine learning methods offer flexibility and can scale with available computational resources, making them well-suited for capturing complex nonlinear dynamics in climate data, classical and semi-structured approaches such as SARIMA and BSTS remain highly competitive for this application, both in terms of short to medium-term prediction and in terms of model interpretability.

1 Introduction

This study evaluates the predictive effectiveness of modern machine learning models against classical temporal models (SARIMA) on global temperatures. Bootstrap resampling and Forward-Chaining Cross-Validation will be utilized in constructing confidence intervals to assess model predictability and stability.

The study will use the Goddard Institute for Space Studies Surface Temperature Analysis (GISTEMP v4) dataset developed by NASA. This dataset contains the averaged land-surface air and sea-surface water temperature anomalies (Land-Ocean Temperature Index, L-OTI), measured in Celsius relative to the 1951 – 1980 baseline.

The dataset includes monthly, seasonal, meteorological annual, and annual means spanning 146 years from 1880 – 2025. The target will focus on the Global Annual Mean (J-D), using monthly data as the explanatory variable. The data contains 19 parameters (including year), and 146 observations, which were later coerced into a series of 1752 monthly observations.

Relative to the original proposal, several adjustments were made to the scope of the project. While the proposal outlined evaluating all models separately across the Northern and Southern Hemispheres, this was not consistently implemented across every method. Instead, hemisphere-specific analysis was incorporated selectively, most notably within the BSTS framework. In addition, BSTS was introduced as an extension beyond the originally proposed set of models, while certain initially considered approaches, such as polynomial regression, were investigated, being either simplified or excluded to maintain a focused and coherent comparison. Overall, the study places greater emphasis on robust model comparison than on exhaustive regional analysis.

1.1 Data

The dataset can be directly exported as 3 separate CVS files (Global, Southern, and Northern Hemispheres). They are very clean and require almost no preprocessing for analysis. The data consisted of only two missing data points. One in the D-J meteorological mean, and the

seasonal (winter) DJF mean for the year 1880. The data is missing because these means take data from the previous year (December 1879); however, as the data begins in 1880 there is no value to complete the calculation. A simple fix would be to average over the data that is present and use that as the 'best guess'. Fortunately, these values were not needed for analysis and can simply be removed.

1.2 Challenges for Temporal Data

By nature, the GISTEMP v4 data for climate anomalies are temporal and highly correlated. As such, the analysis of these data must be conducted in a way that preserves the data's time-based correlation and connectivity. Thus, special consideration was taken into consideration when choosing methodology to address this.

The most notable change is how Cross-Validation should be employed. The usual method of Cross-Validation involves selecting a random sample of data for training and another for testing. However, in temporal data, this style of Cross-Validation introduces data leakage (as the model can see into the future it is attempting to predict), and it destroys the autocorrelation present in the data. Therefore, classical Cross-Validation cannot be used. To account for the temporal dependencies, Forward-Chaining (Rolling-Origin) Cross-Validation with Scikit-learn's `TimeSeriesSplit` was employed. This method maintains the temporal ordering of these data by iteratively moving through a designated training window (lag) for a test set and a forecast (lead) window immediately following. This preserves the chronological autocorrelated structure of the data, creates robust out-of-sample testing, and captures multiple seasonal cycles. A Sliding-Window, not an Expanding-Window (the default), was employed to ensure consistent training size.

In addition, various custom functions were developed for specific model usage. While SARIMA is naturally designed to handle lead and lag data, other models require manually structured lagged data for training. To address this, various custom functions were developed. These will be discussed further in their respective modeling sections.

1.3 Exploratory Data Analysis

Initial analysis of the data revealed significant increases in temperature anomalies from the baseline of 14°C or 57.2°F (average global temperature from 1951-1980 established by Dr. James Hansen). Around 1975 (Figure 1.1), there is a significant increase in global temperatures, reaching a maximum of 1.28 (15.28°C) in 2024. Respectively, the minimum recorded value is -0.49 (13.51°C) occurring in 1909, with an overall average departure of 0.082 (14.082°C) across the entire global data set.

It is also interesting to note that the trends in hemispheres differ over the 146 years of reported data. (Figure 1.2) The Northern Hemisphere is reaching much higher temperatures at 1.65 (15.65°C), while the Southern Hemisphere reaches only 0.91 (14.91°C). There may be several possible reasons for this. Geologically, the southern hemisphere contains significantly more water at around 80%, while the northern hemisphere only boasts 60% water. As the GISTEMP v4 contains averages of both land and sea temperatures, these results likely reflect the thermal inertia of the oceans, where the southern hemisphere’s vast oceans act as a massive heat sink that stabilizes the regional temperatures. Another possibility is the urban heat island (UHI) effect. That is, urban areas tend to see significantly higher temperatures than their neighboring areas. While in 2001, Dr. Hansen made significant efforts to correct this effect by replacing many urban data stations with rural ones, it may still have some effect on the hemispheres, as most of the world’s population lives in the Northern Hemisphere.

Early Autocorrelation Functions and Partial Autocorrelation Functions indicated significant in each of the AR, MA, and Seasonal Components. Augmented Dickey-Fuller (ADF) testing was performed to assess the stationarity of the data. The global monthly data produced a p-value of 0.97, strongly suggesting that first-order differencing was needed to coerce stationarity for the SARIMA baseline modeling in the next section. The corrected data produced an ADF test of 0.01. Diagnostics on the training set indicate that first-order differencing was sufficient to produce stationarity, while first-order seasonal differencing was deemed unnecessary.

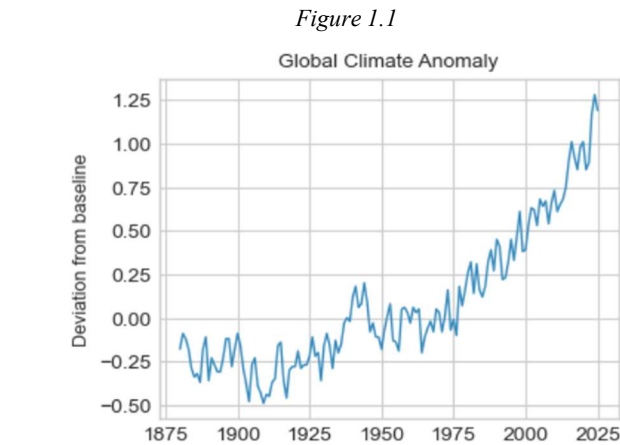
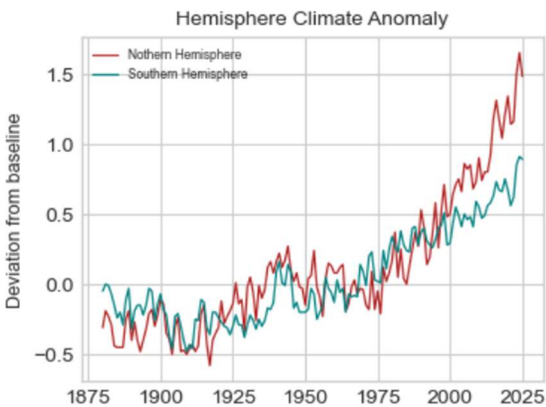


Figure 1.2



2 SARIMA Baseline

SARIMA (Seasonal Autoregressive Integrated Moving Average) modeling was chosen as the classical baseline model for comparison against modern methods. The order of SARIMA(1,1,2)x(2,0,1)x12 (Equation 2.1) with drift was the suggested model using AutoSarima.

Equation 2.1

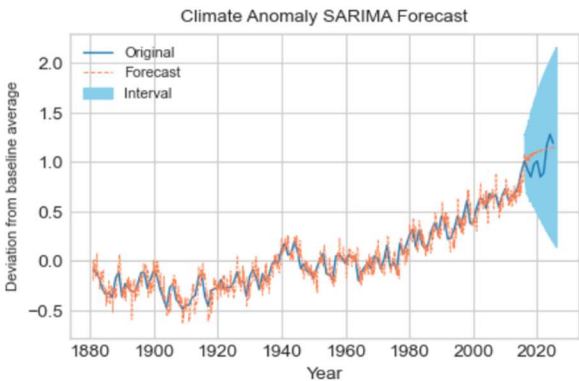
$$(1 - \phi_1 B^1)(1 - \Phi_1 B^{12} - \Phi_2 B^{24})(1 - B)x_t = (1 + \theta_1 B^1 + \theta_2 B^2)(1 + \Theta_1 B^{12})\omega_t + \delta$$

After the 10-year (2015 – 2025) holdout data for testing was removed, the model was trained on the remaining data. The only insignificant coefficient was the intercept with a value of 0.678. However, models without drift struggled to create the overall upward trend present in the data. As this is a common issue in modeling climate data, the drift was kept in the model even though the coefficient was deemed insignificant. The loss and model selection values are shown below, accompanied by the predictive plot. (Table 2.1)

Table 2.1

AIC	BIC	MSE	RMSE	MAE
-2590.3	-2547.2	0.0386	0.1966	0.1729

Figure 2.1



However, the SARIMA model produced a Ljung-Box p-value of 0.04. Recall that the Ljung-Box assumes a Null of white noise in the residuals. Given this p-value, we have reason to believe that SARIMA does not fully capture the trend and autocorrelation available in the data. SARIMA is also notorious for ignoring predictive seasonality by outputting a mean estimate line. The explosion of the interval band may signify instability or overparameterization in the model. These indications

suggest that the SARIMA architecture has reached its limit in capturing the full complexity of the GISTEMP v4 dataset. These results further motivate the use of non-linear Machine Learning methods, aiming for tighter and more robust predictive estimates.

3 Sliding Window (Feed-Forward) Neural Networks:

Two different Neural Networks were employed on the GISTEMP v4 data. Sliding Window Feed-Forward Multilayer Perceptron (FFNN), and the Pinball Mixed Quantile Feed-Forward Network (Pinball).

3.1 Data Handling and Model Architecture

The biggest challenge was adapting this non-linear method to a time-dependent format. To do so, a custom class (*prm*) was created to handle lagged, lead, holdout, and training/testing length. The last 10 years (2015–2025) were removed for holdout final testing. Then the remaining data was standardized using the overall training mean of -0.2424 and standard deviation of 0.154.

For further handling of feeding the lags and leads into the network, another custom function *window* (and later *make_windows_fold*) was developed. The functions design was to input a running set of lagged data for the initial forward pass (24 months), a target lead set (24 months), and then iterate through the entire training data set. This action is akin to a Sliding Window or Feed-Forward Neural Network. A simple multi-layer perceptron architecture was designed for the FFNN and Pinball networks, consisting of 3 hidden layers:

1. *Linear*(24, 16), *LeakyReLU*(), *Dropout*(.2)
2. *Linear*(16, 12), *LeakyRelu*(), *Dropout*(.1)
3. *Linear*(12, 24)

Loss: Huber Loss
Optimizer: Adam

Equation 3.1

$$\text{Huber Loss : } L_{\delta}(a) = \begin{cases} \frac{1}{2}a^2 & \text{for } |a| \leq \delta \\ \delta(|a| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

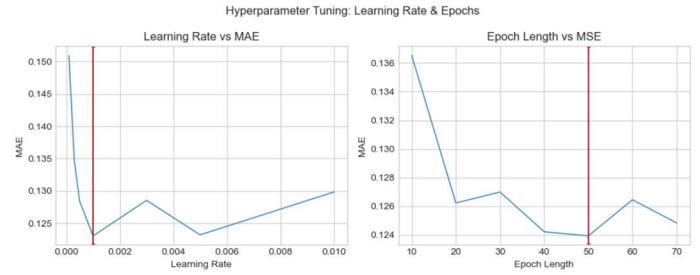
Equation 3.2

$$\text{Leaky ReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha x & \text{if } x \leq 0 \end{cases}$$

The Linear layers were chosen for their simplicity and similarity to the autocorrelation of SARIMA. After experimentation, an autoencoder bottleneck of 24 to 16, then 12, and back to 24 lead outputs was deemed appropriate. Leaky ReLU was chosen for its versatility, speed, and ability to handle the decimal values produced by standardizing without removing neurons. Two layers of dropout were implemented to help correct possible overfitting of the model.

From here, some hyperparameter tuning was conducted to select the most appropriate epoch length and learning rate. (Figure 3.1). The learning rate tended to fluctuate between 0.001 and 0.005, while the number of epochs varied widely. In general, increasing epoch length was better; however, it was not necessarily better in short bursts. That is, several local minima were found, followed by increases in loss. Epochs of 40 to 60 usually landed in nice local minimums (as seen in Figure 3.1) and were used.

Figure 3.1



3.2 Pinball NN for prediction bands

From here, the Pinball Network was developed to produce quantile bands around the Sliding Window Network, providing an advantage of producing a 90% prediction interval. A custom loss function (*pinballHuber*) was created to handle three separate loss values (Equation 3.1). This loss maintains the same Huber point estimate loss, while incorporating a pinball loss for estimating the 5th and 95th quantile estimates. The architecture was similar to the FFNN, with one notable change of $24 \times 3 = 72$ outputs. That is 24 leads for point, 5th-quantile, and 95th-quantile estimates.

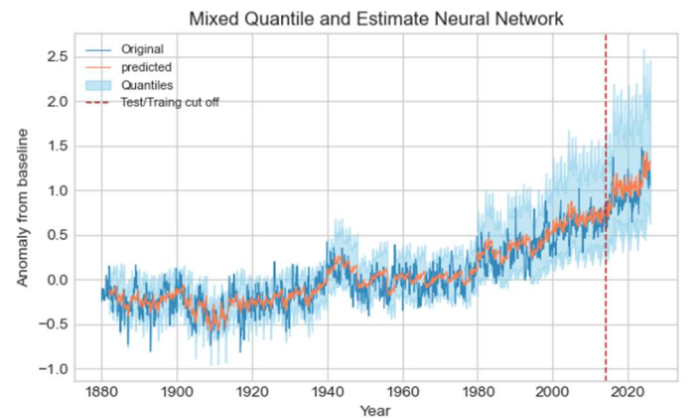
Equation 3.1

$$\mathcal{L}_{Total} = \mathcal{L}_{Huber}(y, \hat{y}_{mid}) + \rho_{1-\tau}(y - \hat{y}_{low}) + \rho_{\tau}(y - \hat{y}_{high})$$

$$\rho_{\tau}(u) = \max(\tau u, (\tau - 1)u)$$

As expected, the Pinball Network performed similarly to the FFNN network in estimates but was more robust by allowing prediction bands. The model also saw slight heteroscedasticity but maintained normalized residuals. (Figure 3.2).

Figure 3.2



3.3 Model Implementation and Limitations

When modeling with the Neural Network, careful consideration was taken to avoid data leakage. To avoid this, the holdout had to be removed before standardizing it. Back-transforming was also implemented based on data missing the holdout values. Further consideration was needed to ensure the Sliding Window did not accidentally leak lead data into the lagged forward passes. Hence, the later implementation of *make_windows_fold* to avoid potential data leakage.

There are a few major concerns about this model. The Network's architecture fixes the output length with *prm.length*. Unfortunately, this

makes forecasting beyond the designed output (24-months in this study) dangerous, and extremely unstable for long-term forecasting. Recursive forecasting must be used for long-term forecasting and was employed to match the 10-year holdout. If the initial forecast is even a little over or underfit, the model will diverge very quickly. When bootstrapping is employed, the long-term forecasting instability is more apparent. (See Figure 7.5) Resetting the output length to the desired forecast length could remove the necessity of recursive forecasting but may hinder the data the model can train on.

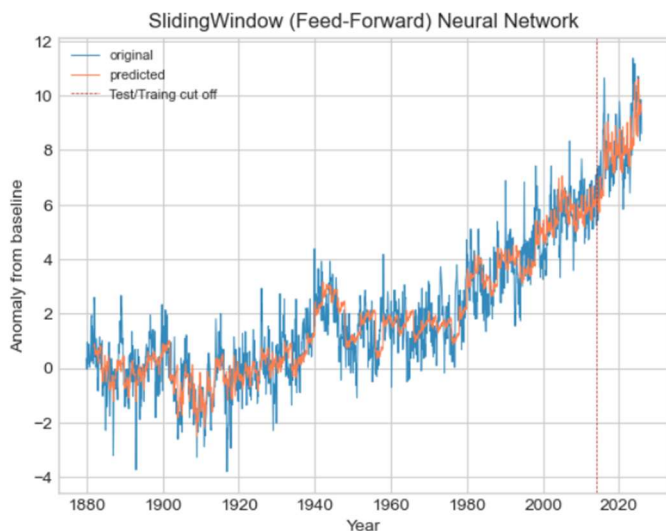
As mentioned above, there are also issues with high variability in the model. Ensemble methods are the next step to reduce this variability. This can be done by running the model several times and averaging predictions. We can also implement a learning rate ‘scheduler’ to slowly reduce or freeze the model to a ‘best’ state, reducing the possibility of overfitting, and allowing finer convergence. However, while this can reduce some stability issues, it will not fix the dangers present in long-term forecasting if recursive forecasting is needed.

For simplicity, the FFNN estimates will be used for model comparison later.

3.4 Observations

For training, 24 months (2 years) of leads were preserved for early MAE/MSE estimates. Following back transforming, the initial FFNN produced an MAE interval of (0.112, 0.139) °C, which was remarkably tight compared to SARIMA. Unfortunately, the model will later suffer from high variance due to strong dependence upon the initial randomized weights, and the need to perform recursive forecasting. However, an early (singular) run of the FFNN produced an MSE of 0.0179 against the Holdout data.

Figure 3.3



The bandwidth on the Pinball has a similar width to that of SARIMA, but the model produced more ‘movement’ in the prediction than SARIMA. The MSE, RMSE, and MAE on the holdout improved over the initial trial as well. The model preforms admirably compared to SARIMA, but SARIMA’s ability to ‘reduce to a mean estimate’ proves advantageous compared to FFNN’s need to use recursive forecasting.

Table 3.1

Model	MSE	RMSE	MAE
FFNN	0.0179	0.1337	0.1041
Pinball	0.019	0.1378	0.123

4 Regularized Regression

A regularized linear autoregressive model was implemented using lagged temperature anomalies (1–24 months), a linear trend term, and seasonal Fourier components (Eq. 7.1). Due to strong collinearity among lag predictors, Elastic Net was used as the primary sparse-shrinkage model (Figure 4.2), with Ridge regression (Figure 4.3) retained as a stability benchmark. Lasso was initially explored but was thought to offer limited benefit in this highly correlated setting and was therefore excluded from the final comparison.

Equation 4.1

$$y_t = \sum_{k=1}^{24} \beta_k y_{t-k} + \gamma t + \delta_1 \sin\left(2\pi \frac{t}{12}\right) + \delta_2 \cos\left(2\pi \frac{t}{12}\right)$$

Despite the strong correlation among lagged predictors, the selected Elastic Net models tended to favor a relatively high L1 contribution, suggesting that some degree of sparsity favored model generalization.

Figure 4.2

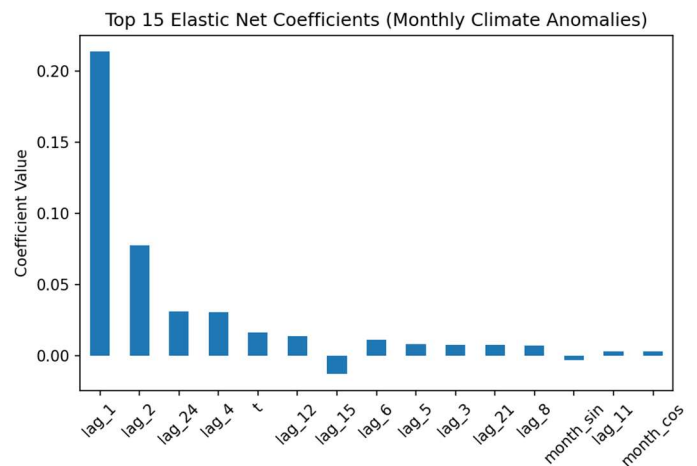
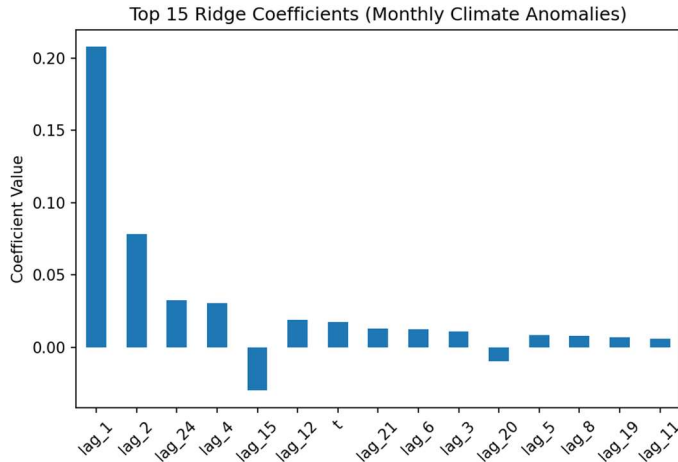
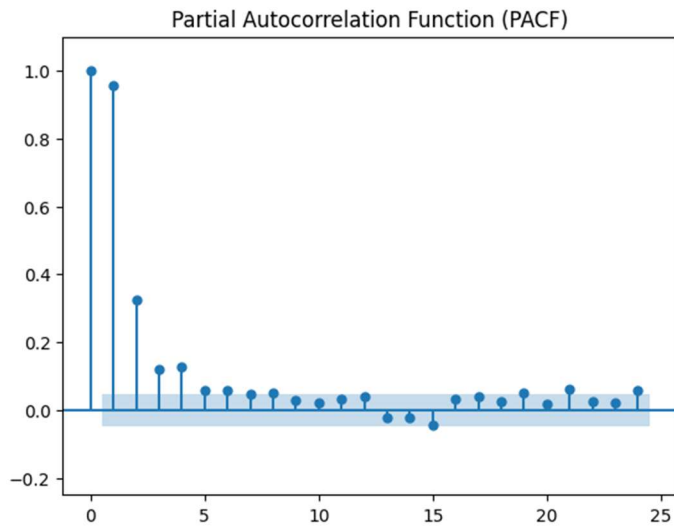


Figure 4.3



While the autocorrelation structure indicates strong persistence across many lags, the partial autocorrelation function indicates that most predictive information is captured by the first 2–4 lags (Figure 4.4). Additional lag terms contribute little additional information, explaining the limited sparsity observed in regularized regression models.

Figure 4.4



4.1 Model Implementation and Limitations

Initial Elastic Net and Ridge models were fit using lagged features constructed from the full time series. While suitable for exploratory analysis, this approach introduced data leakage, as lagged predictors for validation and holdout periods included future observations. This resulted in overly optimistic predictive performance that were clearly following the actual temperature anomalies.

To ensure a fair comparison with other models, the approach was modified as follows:

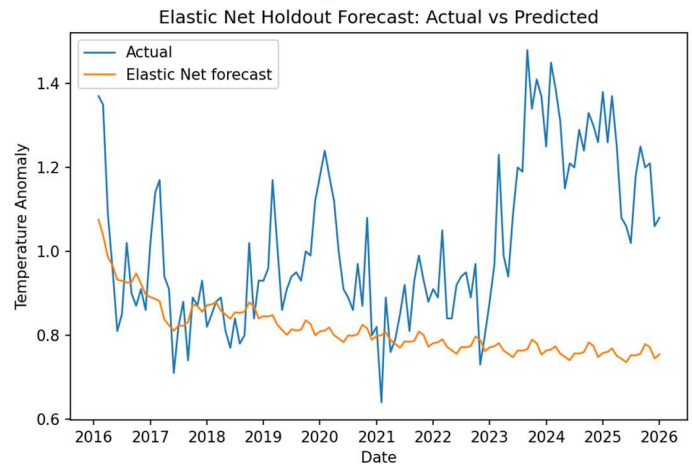
- The series was split into a training set and a final 10-year holdout period
- Models were fit using only pre-holdout data
- Predictions for validation folds and the holdout period were generated recursively, using prior predictions as lag inputs

This recursive forecasting approach prevented leakage by ensuring that no future observations are actually used when constructing lagged predictors.

4.2 Observations

The model performs well for short-term forecasts, capturing persistence in the series through low-order lags. Predictive performance stabilizes quickly, with limited benefit from including higher-order lag terms. Compared to purely statistical models, Elastic Net offers a transparent and interpretable structure but is limited in its ability to capture complex nonlinear dynamics. Recursive forecasting introduces some error accumulation over longer horizons, likely reducing accuracy in the 10-year holdout period, with the model under-predicting the trend (Figure 4.5).

Figure 4.5



5 Bayesian Structural Time Series (BSTS)

The Bayesian Structural Time Series (BSTS) model represents the observed series as a sum of latent components:

Equation 5.1

$$y_t = \mu_t + \gamma_t + \phi_t + \varepsilon_t, \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

where μ_t , γ_t , and ϕ_t represent the trend, seasonal, and autoregressive components, respectively. Each component evolves according to its own stochastic process, allowing the model to flexibly capture both long-term structure and short-term temporal dependence.

The trend component is modeled using a local linear trend specification

Equation 5.2

$$\mu_t = \mu_{t-1} + \beta_{t-1} + \eta_t$$

where μ_t represents the level and β_t the slope, allowing the trend to evolve over time rather than remain fixed.

Inference is performed using variational methods to approximate the posterior distribution of the model parameters.

5.1 Prior Specification and optimization

The BSTS model was implemented using weakly informative priors for all variance and coefficient parameters, as provided by the `sts` module in `tensorflow_probability`. These priors promote numerical stability while allowing the data to inform the model.

Variance-related parameters (observation noise, level evolution, seasonal drift, and autoregressive scale) were assigned LogNormal priors, ensuring positivity and discouraging unrealistically large variance components while maintaining flexibility.

The autoregressive coefficient was assigned a Multivariate Normal prior, centered near zero with symmetric uncertainty. This allows the model to capture either positive or negative temporal dependence.

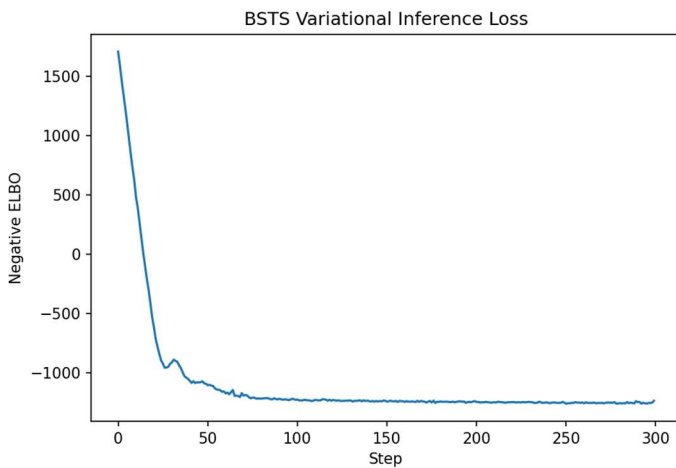
From a machine learning perspective, model fitting was performed using stochastic gradient-based optimization via the Adam optimizer, applied in a variational inference framework. Rather than sampling from the exact posterior (e.g., via MCMC), a surrogate posterior distribution was optimized by minimizing the evidence lower bound (ELBO). This approach enables efficient approximate Bayesian inference, scaling well to larger datasets while retaining uncertainty quantification.

Overall, these priors and optimization choices provide mild regularization and stable, efficient inference without overly constraining the model via the priors.

5.2 Model Implementation and Limitations

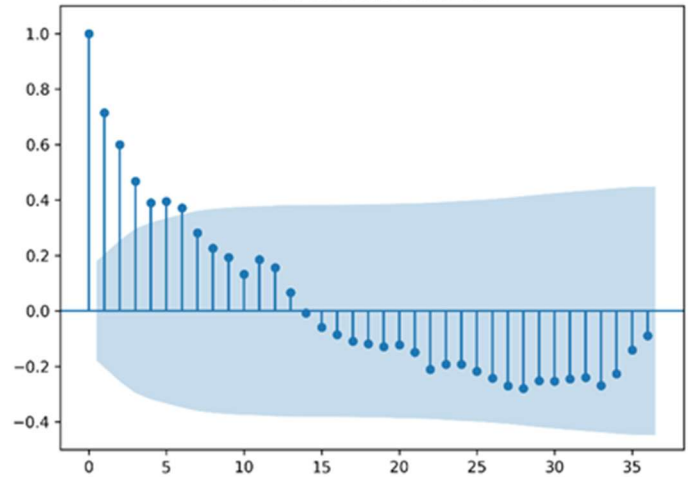
The ELBO diagnostic (Figure 5.1) suggested convergence around ~100 steps; however, in practice, runs below ~150 steps produced unstable results. A value of 200 steps was therefore used to ensure stable convergence.

Figure 5.1



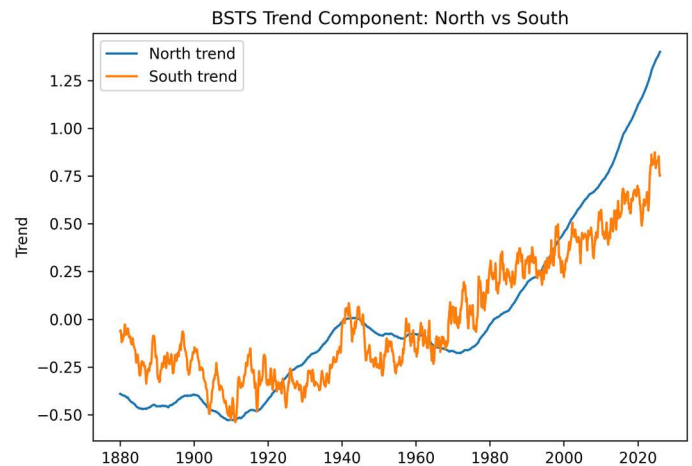
Alternative model specifications were explored: The LocalLevel trend removed long-term structure entirely, leading to underestimation in the holdout period. The LocalLinearTrend (used in the final model) allows a time-varying slope, but can propagate short-term slope fluctuations into long-term forecasts. AR(2) components produced unstable behavior, including unrealistic reversals in long-term trends, whereas AR(1) yielded more stable and accurate predictions. The use of an AR(1) component preserves model simplicity but appears insufficient to capture the full temporal dependence in the data, as evidenced by persistent autocorrelation in forecast errors (Figure 5.2). Predictive performance nevertheless remains comparable to alternative models (refer to model comparison section).

Figure 5.2
ACF of BSTS Forecast Errors



Some instability was observed in the decomposition of latent components, particularly between the Northern and Southern hemisphere models (Figure 5.3). The Northern hemisphere trend was often smoother, likely reflecting better identifiability due to a larger and more informative dataset.

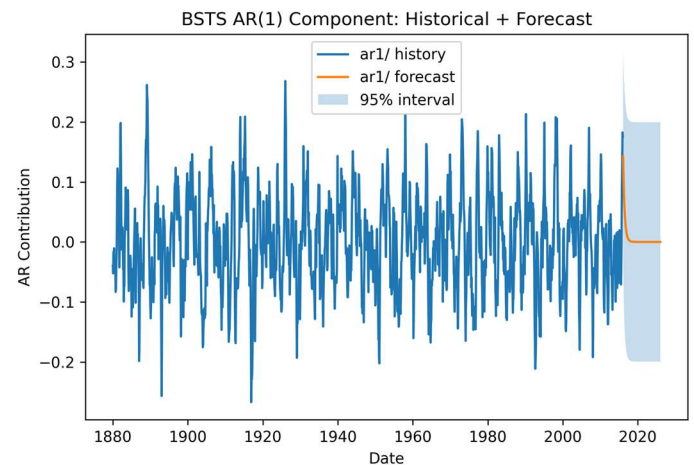
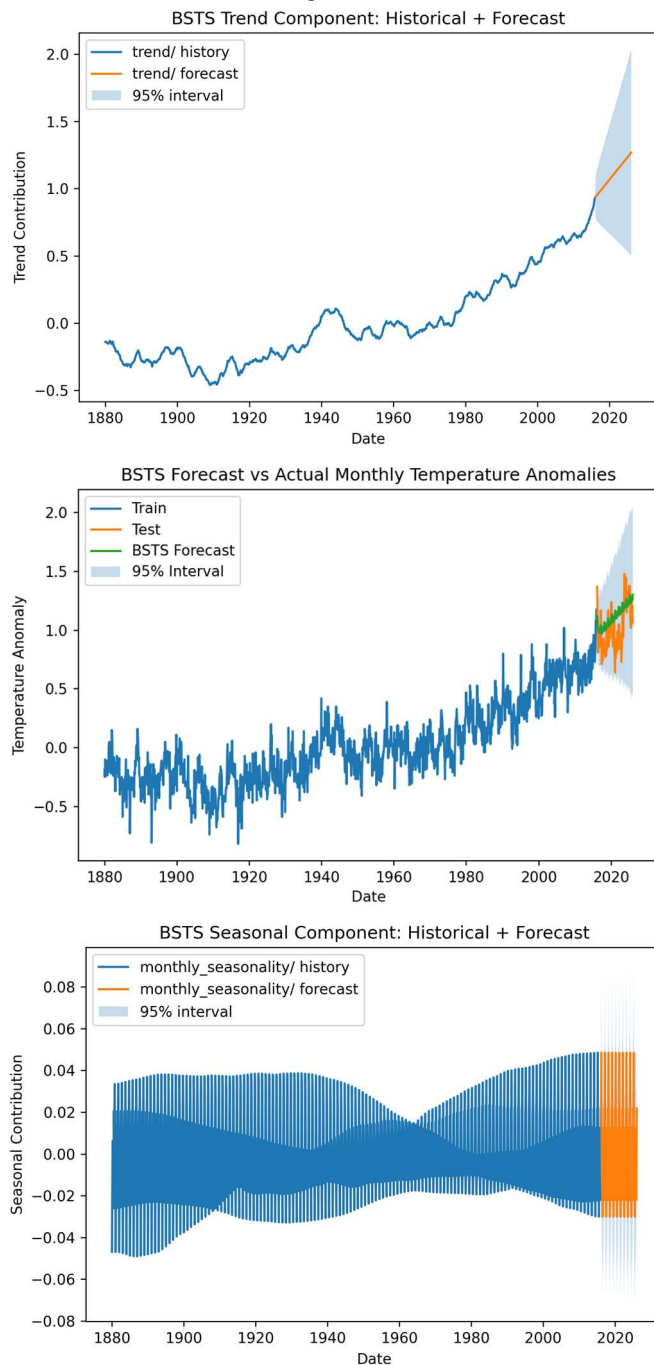
Figure 5.3



Apparent changes in seasonal variability (e.g., reduced seasonality between 1950–1980) appear to be artifacts of the anomaly definition (Figure 5.4), which is normalized to that period.

5.3 Prediction of holdout period forecast

Figure 5.4



5.4 Observations

The model performs well for short-term (~2-year) forecasts, with consistently low cross-validation error. Long-term (~10-year) forecasts tend to slightly overestimate the trend, likely due to sensitivity in the evolving slope component. Some deviations in the holdout period reflect external shocks not captured by the model, such as reduced emissions during the COVID-19 period. The Northern Hemisphere trend is more pronounced and easier to model than the Southern Hemisphere, possibly due to differences in landmass and thermal response. One of the key strengths of BSTS is its interpretability, allowing decomposition into trend, seasonality, and autoregressive components, which are easily visualized and interpreted (Figure 5.4).

6 Random Forest

The Random Forest model was selected as a non-linear ensemble method to capture complex interactions in the climate anomaly data. Unlike the SARIMA baseline, which focuses on linear temporal correlations, the Random Forest regressor builds multiple decision trees to provide a more robust and flexible predictive estimate.

6.1 Model Implementation and Limitations

We fit the model using 24-month lag features. To preserve the chronological structure of the temporal data and avoid data leakage, *Forward-Chaining Cross-Validation* was implemented using Scikit-learn's `TimeSeriesSplit` with a sliding-window approach. The model was optimized via `GridSearchCV` approach, resulting in the best parameters shown in Table 6.1.

A key limitation is that tree-based models like Random Forest are inherently unable to extrapolate beyond the historical range of training data. While these models effectively capture complex non-linear interactions within the training distribution, they may struggle to predict the ongoing upward trajectory of climate anomalies when values exceed the maximums encountered during training.

Table 6.1 Random Forest best parameters from `GridSearchCV`.

Parameter	Value
n_estimators	100
max_depth	None
max_features	None
min_samples_leaf	4

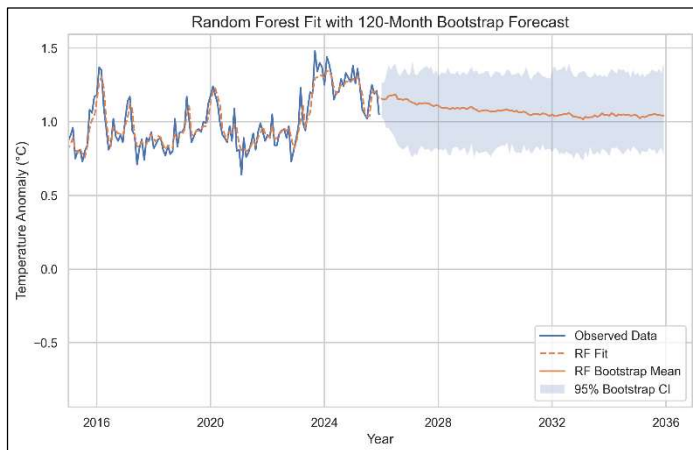
min_samples_split	2
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6.2 Predictions

To generate the 10-year forecast, a recursive forecasting strategy was required. Because the model was trained on 24-month lags, the forecast for month $t + 1$ was used as an input feature for month $t + 2$, and so on. This required iteratively updating the lag feature vector with prior predictions, which can lead to error propagation over long horizons.

In the final model comparison, we utilized bootstrap resampling of the cross-validation residuals to account for the temporal dependence inherent in climate data. The bootstrap results showed only small differences relative to the SARIMA baseline, with the 95% confidence interval for the mean error difference including zero. This indicates that while Random Forest is more flexible and capable of capturing non-linearities, its performance remains competitively stable alongside traditional time-series methods.

Figure 6.1 Random Forest recursive forecast. The forecast captures localized "movement" but does not extrapolate the strong upward trend.



6.3 Observations

The Random Forest model demonstrated strong predictive capabilities compared to the classical SARIMA baseline. The model achieved a mean CV error (MSE) of 0.0216 with a standard deviation of 0.0167. Residuals followed an approximately normal distribution, suggesting the model captured the primary signal. However, the requirement to update lag features led to a slight increase in variance during the holdout period, resulting in some heteroscedasticity.

Unlike SARIMA, which often converges to a smoothed mean estimate for long-term forecasts, the Random Forest predictions maintained "movement," attempting to replicate the local variations found in historical monthly cycles.

7 Model Comparison

The classical baseline model used for comparison was the Seasonal Autoregressive Integrated Moving Average (SARIMA) model. The models included in the comparison were Elastic Net and three machine learning approaches: Neural Network (NN), Random Forest (RF), and Bayesian Structural Time Series (BSTS).

The models were evaluated based on their performance over both long-term and short-term prediction horizons.

Table 7.1 Comparison of cross-validated and holdout performance across models

Model	Mean CV MSE	Holdout RMSE	Holdout MAE	Δ CV vs SARIMA	95% CI (Δ CV)
SARIMA	0.019	0.203	0.177	0	—
Random Forest	0.022	0.194	0.163	0.002	[-0.003, 0.011]
BSTS	0.025	0.184	0.154	0.005	[-0.003, 0.017]
Elastic Net	0.028	0.292	0.222	0.009	[0.001, 0.017]
Sliding Window NN	0.045	0.166	0.133	0.026	[0.003, 0.064]

7.1 Long-term Forecast Performance

Model performance was evaluated using predictions over the final 10-year holdout period (Figure 7.1-Figure 7.2).

BSTS performs best overall, capturing the upward trend with moderate variation and yielding the lowest residual standard deviation. SARIMA provides a solid baseline, capturing the general trend but producing overly smooth estimates that miss peaks. Elastic Net exhibits a downward bias and fails to capture the recent upward trend. Random Forest produces near-constant predictions and is unable to extrapolate beyond the training mean. The Sliding Window neural network shows strong holdout performance, however, this performance is unstable in practice due to recursive error amplification, as each two-year forecast is used as input for subsequent predictions, often leading to compound deviations.

Figure 7.1

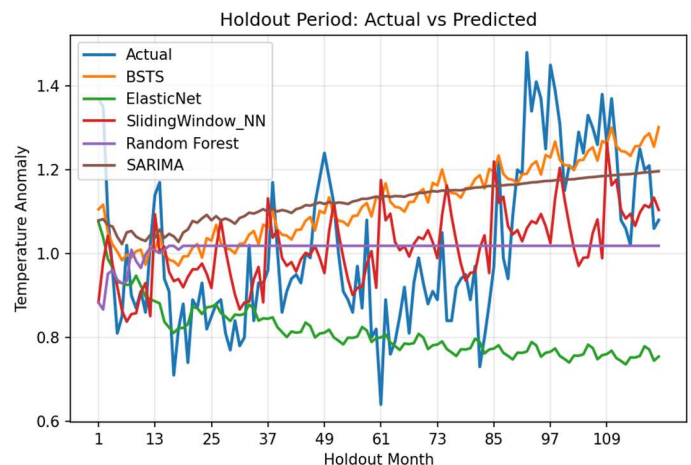
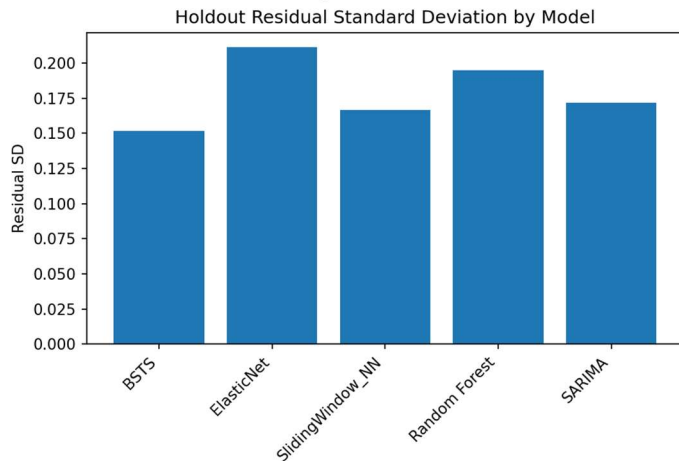


Figure 7.2



7.2 Short-term Forecast Performance (Cross-Validation)

Cross-validation (CV) predictions were assessed over a rolling two-year window using a time-series-aware k-fold cross-validation scheme. This approach ensured that each method was trained and validated on identical chronological data partitions. Minor differences in effective training samples due to the 24-month burn-in period required by lag-based models are discussed in the limitations section below.

Model performance was evaluated using 5-fold time series cross-validation (Figure 7.3-Figure 7.4). SARIMA performed best overall, achieving the lowest mean cross-validation error and demonstrating stability across folds. Random Forest was competitive, with low mean error and moderate variability. BSTS showed moderate performance, with slightly higher error but relatively stable results. The Sliding Window neural network was inconsistent, exhibiting higher error and variability across folds. Elastic Net performed worst, with the highest error and the greatest fold-to-fold variability.

Figure 7.3

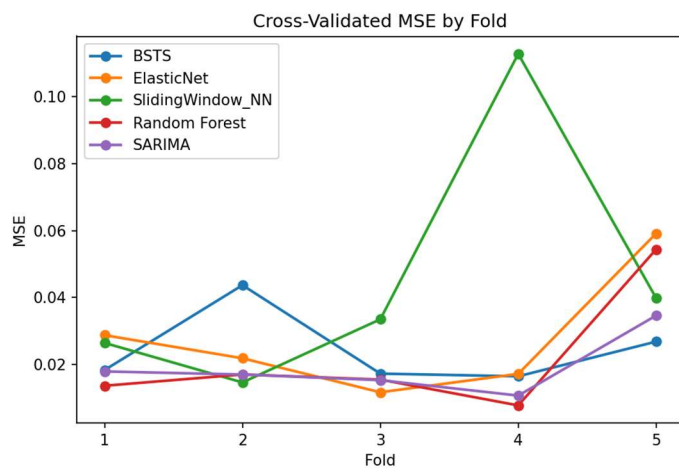
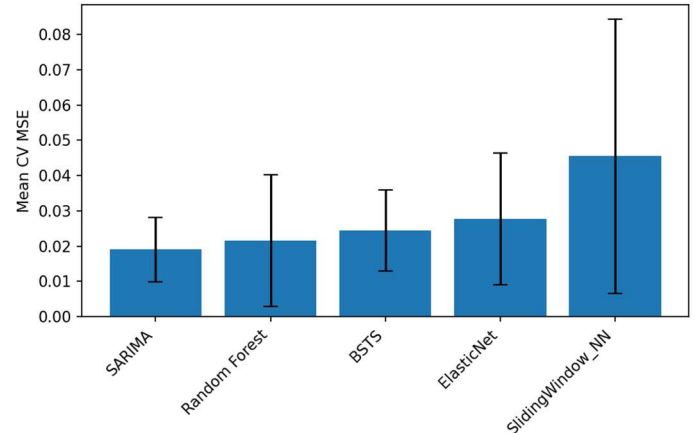


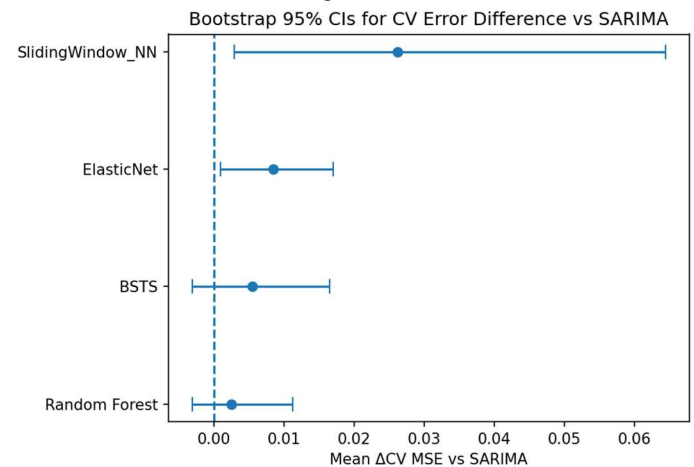
Figure 7.4

Mean Cross-Validated MSE with Fold-to-Fold SD



Statistical comparison versus SARIMA using bootstrap confidence intervals is shown in (Figure 7.5). BSTS and Random Forest show small differences relative to SARIMA, with confidence intervals that include or are near zero. The Sliding Window neural network performs worse than SARIMA, with the confidence interval shifted in the positive direction. Elastic Net performs significantly worse, with a large positive difference and a wide confidence interval.

Figure 7.5



7.3 Model Comparison Limitations

Lag construction and burn-in represent an inherent limitation of lag-based models, which require an initial period to construct features and therefore result in a slightly reduced effective training sample. Given the length of the dataset and the focus on recent performance (final 20 years), this reduction is not expected to materially impact model comparisons.

Model evaluation over a 24-month forecast horizon also introduces differences in forecasting approach across methods. SARIMA and BSTS were fit on training folds and used to generate direct multi-step forecasts from the fitted models. In contrast, Elastic Net was fit on lagged features from the training folds and used to generate forecasts recursively over the horizon, while the Sliding Window neural network was fit on training windows and produced multi-step forecasts iteratively from prior predictions.

These differences in forecasting strategy, particularly direct versus recursive approaches, may affect comparability. Recursive methods, such as Elastic Net and the neural network, are more susceptible to error

accumulation across the forecast horizon, which could partially explain their reduced performance relative to SARIMA and BSTS.

8 Conclusion

This study compared classical and machine learning approaches for forecasting global temperature anomalies using the GISTEMP v4 dataset, taking into account the temporal structure in designing comparable evaluation frameworks. By implementing time-series aware cross-validation and a 10-year holdout period, we ensured a robust and fair evaluation between models.

Results indicate that SARIMA acts as a strong and stable baseline for comparison, particularly for short-term forecasting. BSTS provides the best long-term performance, effectively capturing the upward trend, while yielding significant advantages in interpretability, but it introduces additional complexity and can be sensitive to model specification, prior choices, and optimization settings. Random Forest demonstrated competitive short-term accuracy but lacked the ability to extrapolate beyond the training range. The sliding window neural network exhibited high variability and instability in long-horizon forecasts, while Elastic Net underperformed due to its limited ability to capture nonlinear dynamics and long-term trends.

Overall, these findings highlight that while machine learning methods offer flexibility and scalability, their performance in time series forecasting depends critically on choices in model design and evaluation strategy. For climate anomaly prediction, structured approaches such as SARIMA and BSTS remain highly competitive, particularly when robustness and interpretability are essential, with SARIMA retaining a significant advantage in model simplicity and stability.

9 Societal Impact

Climate forecasting models such as those evaluated in this study have direct implications for policy, infrastructure planning, and public communication of climate risk. The choice of modeling approach directly impacts how uncertainty is quantified and how information is conveyed. Machine learning models are highly flexible and scalable, and are competitive with classical models in predictive power, but may be subject to reduced interpretability, which may directly impact communicability of the findings, having a direct impact on public policy. Structured models such as SARIMA and BSTS offer more transparent representations of trend and variability, which may be more suitable in a public policy context, allowing for rapid and clear communication of critical information. Model instability and variability, particularly in recursive forecasting approaches, highlights the importance of including error bounds and highlighting model limitations in any public forum. The importance of quantifying uncertainty and sharing these limitations is amplified by a lack of public trust in the domain of climate science, where overconfidence in predictions may be a recipe for aggravating this dynamic.

10 Extensions and Improvements

The Sliding-Window Neural Network can be significantly enhanced by implementing a wrapper function that accepts parameterized training and output lengths. This architecture allows better control over the forecast horizon and will effectively remove the need to perform recursive forecasting, reducing cumulative error. Furthermore, model

stability can be improved through the integration of a learning rate scheduler, and ensemble averaging, aggregating weights on k-runs.

A natural extension of this study is to apply the frameworks to regional data. Preliminary analysis suggests that temperature anomalies vary by hemisphere; however, it remains unclear if the spatial dependency persists at a regional scale. By deploying the discussed models, we can critically evaluate whether district climate deviations exist between geographically diverse regions.

11 Team Contributions & Collaboration

Section	Main Contributor	
	Code	Report
Abstract and Conclusion	----	Brendan
Introduction and Data	----	Dustin/Brendan
Data Processing	All	All
EDA	Dustin	Dustin
SARIMA	Dustin	Dustin
Neural Network	Dustin	Dustin
Regularized Regression	Brendan	Brendan
BSTS	Brendan	Brendan
Random Forest	Cesar	Cesar
Model Comparison	Brendan	Brendan